

22-11-2016
BILKENT UNIVERSITY
Department of Electrical and Electronics Engineering
EEE211 Analog Electronics
Midterm Exam

SOLUTION

Surname: _____

Name: _____

ID-Number: _____

Signature: _____

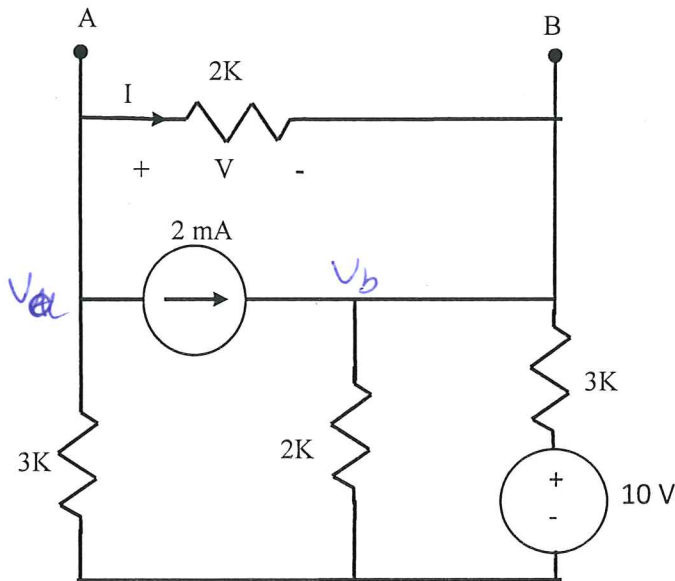
Duration is 110 minutes. There are 4 questions of equal weight, solve all.
Show all your work. No books or notes.

Q1	
Q2	
Q3	
Q4	
Total	

Q1)

a) (15 points) Find V and I in the following circuit.

b) (10 points) Find the Thevenin resistance seen from the terminal defined by "A" and "B".



$$a) \frac{V_a}{3} + 2 + \frac{V_a - V_b}{2} = 0$$

$$2V_a + 12 + 3V_a - 3V_b = 0$$

$$\boxed{5V_a - 3V_b = -12}$$

$$-2 + \frac{V_b}{2} + \frac{V_b - 10}{3} + \frac{V_b - V_a}{2} = 0$$

$$-12 + 3V_b + 2V_b - 20 + 3V_b - 3V_a = 0$$

$$\boxed{8V_b - 3V_a = 32}$$

$$\begin{cases} 15V_a - 9V_b = -36 \\ 40V_b - 15V_a = 160 \end{cases}$$

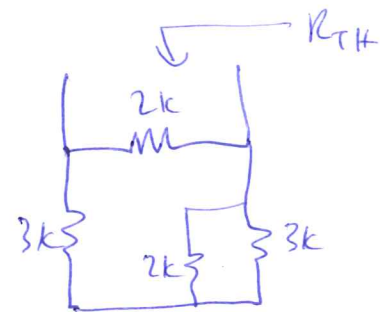
$$\rightarrow 31V_b = 124 \quad \boxed{V_b = 4}$$

$$5V_a - 12 = -12 \quad \boxed{V_a = 0}$$

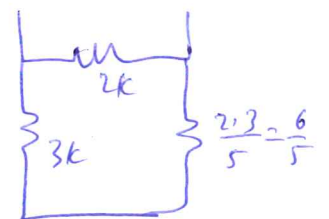
$$\Rightarrow \boxed{V = -4}$$

$$\boxed{I = -2 \text{ mA}}$$

b)



|||



$$R_{TH} = 2 \parallel \left(3 + \frac{6}{5} \right)$$

$$= 2 \parallel \frac{21}{5}$$

$$= \frac{2 \times \frac{21}{5}}{2 + \frac{21}{5}}$$

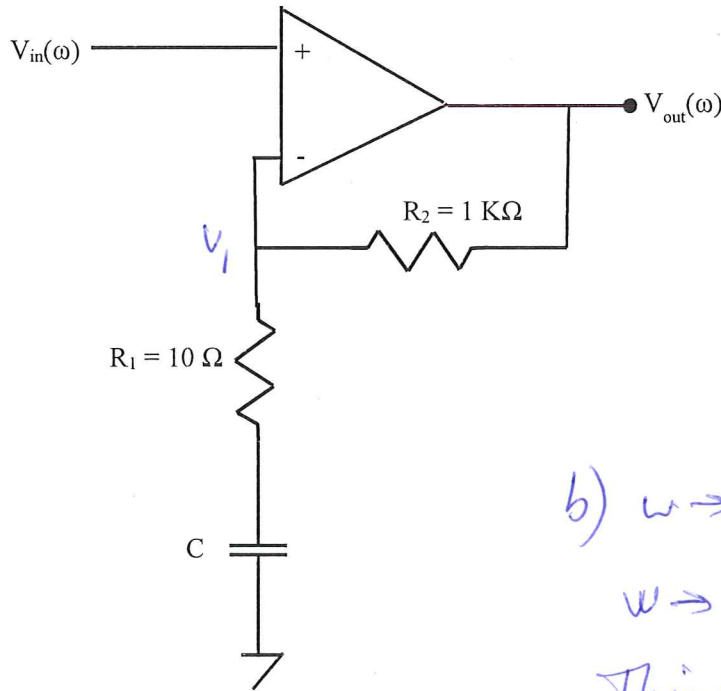
$$= \frac{2 \times 21}{31} = \frac{42}{31} \text{ k}\Omega$$

Q2)

a) (13 points) Derive and find $H(\omega) = V_{out}(\omega)/V_{in}(\omega)$

b) (6 points) What are the voltage gains of this circuit at very low and very high frequencies? What kind of a filter is it?

c) (6 points) Find the value of C for which the -3dB cut-off frequency is 3KHz.



b) $\omega \rightarrow 0 \Rightarrow \frac{V_{out}}{V_{in}} \rightarrow 1$
 $\omega \rightarrow \infty \Rightarrow \frac{V_{out}}{V_{in}} \rightarrow \frac{1010}{10} = 101$
 This is a high pass filter

a) $V_i = V_{in}$

$$\frac{V_{in}}{10 + \frac{1}{j\omega C}} = \frac{V_{out} - V_{in}}{1000}$$

$$V_{out} - V_{in} = \frac{1000}{10 + \frac{1}{j\omega C}} V_{in}$$

$$V_{out} = V_{in} \left(1 + \frac{1000j\omega C}{1 + j\omega C \times 10} \right)$$

$$\frac{V_{out}}{V_{in}} = \frac{1 + j\omega C \times 10 + j\omega C \times 1000}{1 + j\omega C \times 10}$$

$$= \frac{1 + j\omega C (1010)}{1 + j\omega C \times 10}$$

c) $\frac{1 + \omega^2 C^2 \times 1010^2}{1 + \omega^2 C^2 \times 10^2} = \frac{101^2}{2}$

$$2 + \omega^2 C^2 \times 2 \times 1010^2 = 101^2 + \omega^2 C^2 \times 10^2 \times 101^2$$

$$\omega^2 C^2 (200 \times 101^2 - 100 \times 101^2) = 101^2 - 2$$

$$\omega^2 C^2 \times 100 \times 101^2 \approx 101^2$$

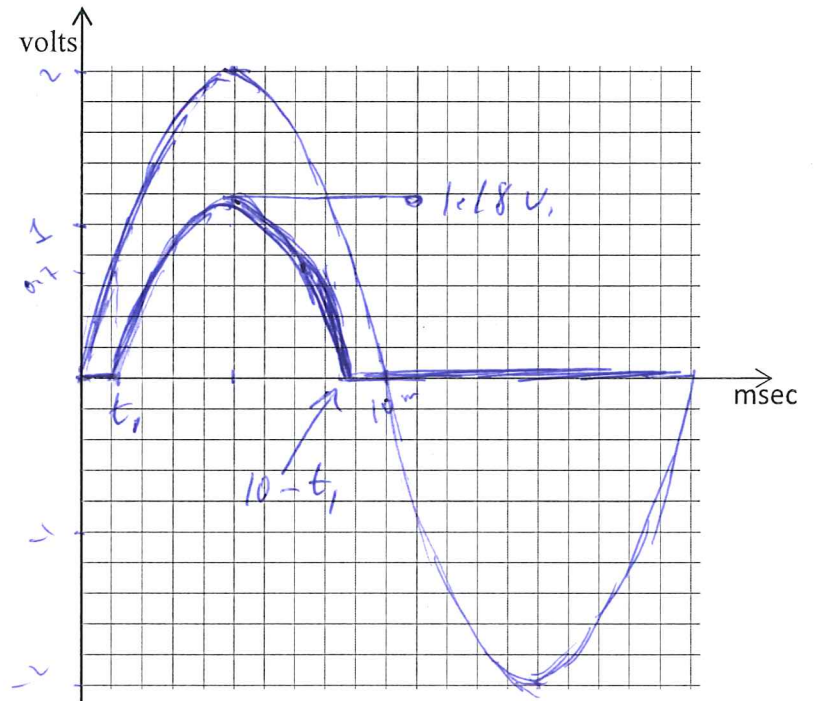
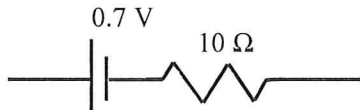
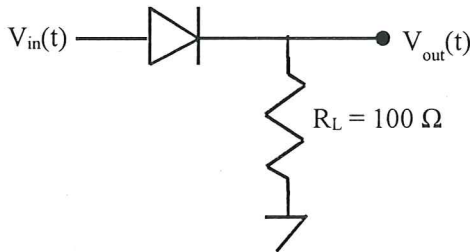
$$\omega^2 C^2 = \frac{1}{100}$$

$$\omega C = \frac{1}{10} \quad C = \frac{1}{10 \omega}$$

$$C = \frac{1}{10 \times 2\pi \times 3000}$$

$$= 5.3 \mu F$$

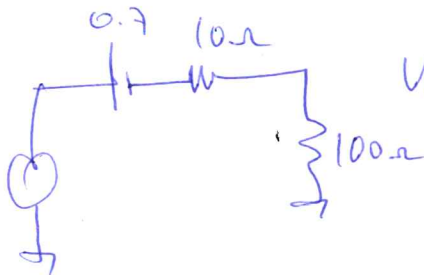
Q3) (25 points) For the diode circuit below and for $V_{in}(t) = 2\sin(2\pi 50t)$ volts, where t is time in seconds, draw the input and output waveforms, $V_{in}(t)$ and $V_{out}(t)$, on the chart given below. The diode in the circuit has the model shown below (0.7 V in series with $10\ \Omega$) when it is ON. Scale the chart with the given units: volts and msec. Find and indicate the critical times and values.



The diode is OFF until $2\sin\omega t_1 = 0.7$

$$\Rightarrow t_1 = \frac{\sin^{-1}(0.35)}{2\pi \times 50} = 1.1 \text{ msec}$$

When diode is ON:

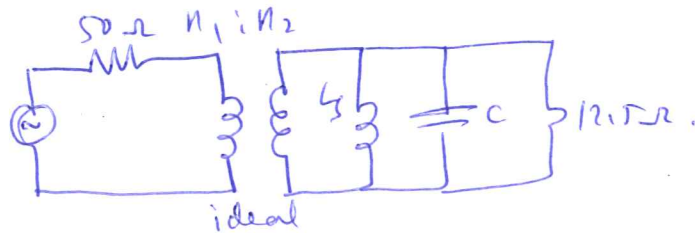


$$V_{out} = \frac{2\sin\omega t - 0.7}{100 + 10} \times 100 \quad t_1 \leq t \leq 10 - t_1$$

$$V_{out\text{peak}} = \frac{2 - 0.7}{110} \times 100 = 1.18 \text{ V}$$

Q4) (25 points) A voltage source with 50Ω internal (source) resistance is to be used to deliver power to a load which consists of a resistor ($= 12.5 \Omega$) in parallel with a capacitor $C = 10 \mu\text{F}$.

Design a transformer to connect between the source and the load so that maximum power is delivered to the load at $\omega = 10^8 \text{ rad/s}$. For the transformer we have a toroid with $A_L = 100 \text{ nH/T}^2$.



Tune out C using L_s

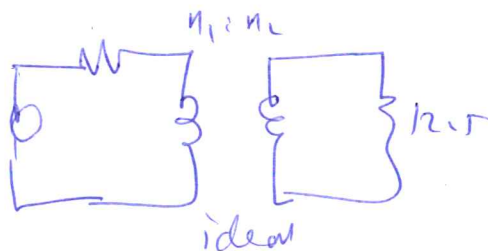
$$L_s = \frac{1}{\omega^2 C} = \frac{1}{10^{16} \times 10 \times 10^{-12}} = 10^{-7} \text{ H} = 10 \mu\text{H}$$

$$L_s = n_2^2 A_L$$

$$n_2^2 = \frac{L_s}{A_L} = \frac{10 \mu\text{H}}{100 \text{ nH/T}^2} = \frac{10^{-5}}{10^{-7}} = 10^2 = 100$$

$$n_2 = \sqrt{100} = 10$$

Thus we have



For max pow. tr. $50 = \left(\frac{n_1}{n_2}\right)^2 \times 12.5$

$$\left(\frac{n_1}{n_2}\right)^2 = \frac{50}{12.5} = 4$$

$$\frac{n_1}{n_2} = 2 \quad n_1 = 2n_2 = 20$$

$$\boxed{\begin{matrix} n_1 = 20 \\ n_2 = 10 \end{matrix}}$$